

Lecture 4 (9/21)

Continuous Distributions

The uniform distribution

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$$P(y_1 \leq Y \leq y_2) = \frac{y_2 - y_1}{b - a}.$$

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In code

```
y = np.random.uniform(a, b, size=1000)
```

Example

The time I wait at the bus stop in minutes is $T \sim \text{Uniform}(0, 5)$ (not very realistic).

- What is the probability I wait more than 2 minutes?
- What is the average time I wait, given that I wait longer than 3 minutes?

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Example

Consider

$$\begin{aligned} X &\sim \text{Bernoulli}(1/4) \\ Y|X &\sim \text{Uniform}(0, 1 + X) \end{aligned}$$

- Compute (i) $E[Y]$ and (ii) $P(Y = 1|X > 1/2)$.
- Translate this problem into code and estimate these from samples.

Example

We often use $\text{Uniform}(0, 1)$ to model an unknown probability. Say Y is the outcome of a binary election. Before we do any polling we have complete ignorance about the result. We model this as

$$\begin{aligned} X &\sim \text{Uniform}(0, 1) \\ Y|X &\sim \text{Bernoulli}(X) \end{aligned}$$

Densities

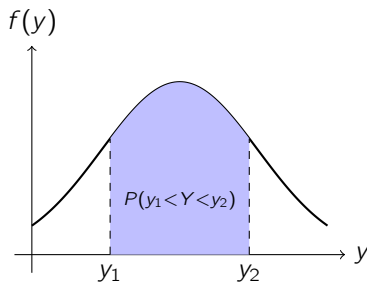
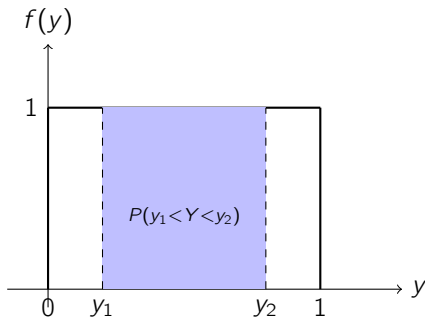
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Densities

A continuous random variable Y has a density $f(y) \geq 0$ with

$$P(a < Y < b) = \int_a^b f(y) dy, \quad \int_{-\infty}^{\infty} f(y) dy = 1.$$

For the uniform case, $f(y) = \frac{1}{b-a}$ on $[a, b]$. Note $f(y)$ can exceed 1; the only requirement is that the total area under f equals 1.

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- Often we use continuous random variables to model things like grades, or the number of people, because from a mathematical perspective they can be simpler to work with.
- The most important thing for this class is to understand conceptually how probability calculations work, not to evaluate the integrals to get exact results.

Density vs. CDF

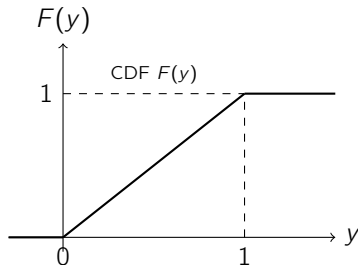
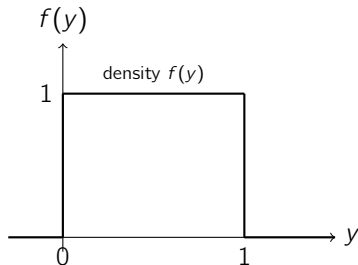
- The density $f(y)$ tells us how probability is spread out: probabilities come from *areas* under f .
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$$F(y) = P(Y \leq y) = \int_{-\infty}^y f(t) dt \implies f(y) = F'(y)$$

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Example

Let $U \sim \text{Uniform}(0, 1)$ and find the CDF and densities of U^2 and $\sqrt{U}$.

Example

Let $U \sim \text{Uniform}(0, 1)$ and define $T = -\frac{1}{\lambda} \log(U)$. Find the CDF and density of T .